

Checking the $+$ –prescription split

Two things right, one leftover to delete, and a divergence in the first term

$$\kappa \equiv k \cdot (y' - z), \quad \xi \equiv p^+ / k^+, \quad \Xi \equiv (\vee - k^+) / k^+, \quad \ell_k \equiv \log \frac{k^+}{\Lambda}$$

0. Verdict

item	status
sign of the second term (+, from the – in the bracket against the overall –)	correct
prefactor $\frac{1}{8\pi^5} = \frac{1}{4\pi^4} \cdot \frac{1}{2\pi}$	correct
numerator $(y' - z) \cdot (x - z) (x' - y') \cdot (y - w)$ — the $\delta_{jm} \delta_{ik'}$ contraction	correct
phase dropped in the second term ($g(0)$ has phase 1)	correct
the bracket still standing in the second term	delete it (§1)
“the first term has no transverse divergence”	false (§2)
$\vee \rightarrow \infty$ behaviour of each piece	spurious log $\vee$ (§3)

1. The leftover bracket

The second term carries $\left[\frac{\delta_{k'm} \delta_{ij}}{p^+ + k^+} - \frac{\delta_{jm} \delta_{ik'}}{p^+} - \frac{\delta_{im} \delta_{jk'}}{k^+} \right]$. It should not. Three reasons, any one sufficient:

- the contraction has **already been done** — it is sitting in the numerator you wrote, $\delta_{jm} \delta_{ik'}$ on $(y' - z)^m (x' - y')^{k'} (y - w)^i (x - z)^j$ giving $m=j : (y' - z) \cdot (x - z)$ and $i=k' : (x' - y') \cdot (y - w)$;
- the $\frac{1}{p^+}$ has **already been integrated** — it is the $\log \frac{\vee - k^+}{\Lambda}$;
- there is no $\int dp^+$ left, so a free p^+ in that term is undefined.

The second term should read

$$+ \frac{1}{(2\pi)^3} \frac{ig^4 f^{cd'a}}{8\pi^5} \frac{1}{k^+} \log \left[\frac{\vee - k^+}{\Lambda} \right] \int_{w, y', z, y, x, x'} e^{-ik \cdot (y' - w)} \frac{(y' - z) \cdot (x - z) (x' - y') \cdot (y - w)}{(y' - z)^2 (x' - y')^2 (y - w)^2 (x - z)^2} \times [\text{colour}].$$

2. The first term is transversely divergent

Write the three index structures by their z –dependence, with $a = x' - y'$ and $b = y - w$ the spectator vectors:

$$S_1 = \frac{(y' - z) \cdot a (x - z) \cdot b}{(y' - z)^2 (x - z)^2}, \quad S_2 = \frac{(y' - z) \cdot (x - z)}{(y' - z)^2 (x - z)^2}, \quad S_3 = \frac{(y' - z) \cdot b (x - z) \cdot a}{(y' - z)^2 (x - z)^2}.$$

All three fall only as $1/z^2$, with nonvanishing angular average $\rho^2 \langle S_1 \rangle, \rho^2 \langle S_3 \rangle \rightarrow \frac{a \cdot b}{2}$ and $\rho^2 \langle S_2 \rangle \rightarrow 1$:

rho	S1 (d_k'm d_ij)	S2 (d_jm d_ik')	S3 (d_im d_jk')
1e+02	-2.720213	1.000230	-2.720317
1e+03	-2.719644	1.000002	-2.719645
1e+04	-2.719638	1.000000	-2.719638
predicted	(x' - y') \cdot (y - w) / 2 = -2.719638	1	(y - w) \cdot (x' - y') / 2 = -2.719638

so *every* structure is logarithmically divergent in z **unless a phase cuts it**.

Which ones keep their phase

S_1 and S_3 sit on $\frac{1}{p^+ + k^+}$ and $\frac{1}{k^+}$, which are untouched by the prescription, so they keep the full $e^{-i\xi k \cdot (y' - z)}$. The oscillation kills the angular average and their z integrals converge for every $\xi > 0$, leaving $\log \frac{1}{\xi}$; and neither coefficient has a $\frac{1}{\xi}$ pole, so $\int_0 d\xi \log \frac{1}{\xi}$ converges. **Those two are fine.**

S_2 is **not**. The prescription replaces the phase by $(e^{-i\xi k \cdot (y'-z)} - 1)$, and the -1 has *no phase at all*. What is left under $\int d^2z$ is S_2 bare — and S_2 is exactly the **BK real-emission kernel** $\mathcal{K}(x, y'; z)$, which falls as $1/z^2$. Measured $\rho^2 | \langle (e^{-i\xi k} - 1) S_2 \rangle |$ does **not** decay:

xi = 0.3	rho=1e+02: 0.865102	rho=1e+03: 1.043748	rho=1e+04: 1.006886
xi = 0.05	rho=1e+02: 1.038685	rho=1e+03: 1.107809	rho=1e+04: 0.967298

compared with $|\langle e^{-i\xi k} S_2 \rangle| \rho^2 = 1.4 \times 10^{-1}, 4.4 \times 10^{-2}, 6.9 \times 10^{-3}$ for the unsubtracted phase at $\xi = 0.3$. So $\int \rho d\rho (-1/\rho^2) = -\log$: **the first term diverges logarithmically at large $|z|$, for every value of p^+** .

And the same divergence sits in your second term — it is $g(0)$, whose z integral is the same bare $\mathcal{K}$. **The two divergences cancel only in the sum**, i.e. only if you never split. That is the concrete form of the warning from last time: the $+$ prescription moves the transverse divergence, it does not remove it. Splitting a finite object into two infinite ones is not progress.

3. A second, independent problem: the upper limit

The unsplit p^+ integral has a finite $\vee \rightarrow \infty$ limit. The split does not: the subtracted piece is $\text{Ei}(-i\kappa\Xi) - \gamma_E - \log(i\kappa\Xi) \rightarrow -\log\Xi$, and the log piece is $+\log\frac{\Xi}{\lambda}$. Each slides with $\vee$; only the sum is stable:

Ξ	unsplit	+ piece	$\log(\Xi/\lambda)$
10^2	$19.61743 - 1.56527i$	$-5.71101 - 4.70686i$	25.3284
10^4	$19.61538 - 1.57084i$	$-10.31823 - 4.71243i$	29.9336
10^6	$19.61542 - 1.57080i$	$-14.92335 - 4.71239i$	34.5388

Fix: subtract only below k^+

$$\left[\frac{1}{p^+} \right]_+^{(k^+)} g \equiv \frac{g(p^+) - \theta(k^+ - p^+) g(0)}{p^+}, \quad \text{isolating } \ell_k = \log \frac{k^+}{\Lambda} \text{ instead of } \log \frac{\vee - k^+}{\Lambda}.$$

Now the $+$ piece is $\vee$ -stable, and the isolated log is the one BK evolution wants:

Xi=1e2:	+ piece with theta cut =	-1.105837 - 4.706860i
Xi=1e4:		-1.107888 - 4.712428i
Xi=1e6:		-1.107844 - 4.712389i

This also settles the scheme point from the previous note: the finite difference $\log \frac{\vee - k^+}{k^+}$ is now explicitly where it belongs, in the impact factor.

4. What to do

1. **Delete the bracket** from the second term.
2. **Collect the virtual diagrams first**. They replace $-2\mathcal{K}$ by $\mathcal{M}(x, y'; z) = \frac{(x-y')^2}{(x-z)^2(y'-z)^2} = \frac{1}{(x-z)^2} + \frac{1}{(y'-z)^2} - 2\mathcal{K}$, whose large- ρ angular average vanishes at order ρ^{-2} ($1+1-2=0$) so that it falls as ρ^{-4} . Only then is $g(0)$ transversely finite and the split legitimate.
3. **Use the $\theta(k^+ - p^+)$ subtraction** so each piece has a $\vee \rightarrow \infty$ limit and the isolated log is ℓ_k .
4. Then the second term is $\ell_k \times \mathcal{M} \otimes (\text{LO})$, to be absorbed into the rapidity evolution of the Wilson-line correlators, and the first term is the NLO impact factor — finite in both p^+ and the transverse integrals.

A useful self-check at step 4: with the virtuals in, the answer must carry a **single** ℓ_k . If you still find ℓ_k^2 , the subtraction term is still transversely divergent and something upstream is missing — the double log is the signature of exactly the entanglement diagnosed in §2.

zint/src/plus_split_check.py — every number above, written to zint/RESULTS_plusplit.txt.